

Activity Overview

During Phase 3, our efforts centered on three key areas: (1) expanding the training corpus, (2) establishing a robust benchmark for evaluating generalizability, and (3) developing a novel deep learning architecture capable of modeling the sequential structure of speech. To increase the diversity and representativeness of the training data, while addressing bias observed in Phase 2, we downloaded the complete DementiaBank repository. Following a systematic inspection and filtering process, we excluded specific studies (see Data). Building upon the audio preprocessing pipeline developed during Phase 2, we adapted and extended it to accommodate longer recordings without compromising signal integrity or processing consistency across variable durations (see Data).

To enable rigorous assessment of model generalization and bias, we implemented a new benchmark. This setup included standardized data partitioning strategies and an evaluation protocol optimized for out-of-distribution performance analysis, along with additional metrics aligned with real-world clinical applicability (see Generalizability). For model development, we refined our Phase 2 architecture by transitioning from distilled Whisper to Whisper Large V3. Recognizing that Whisper processes inputs in ~30-second spectrogram segments, we repurposed the encoder to act as a feature extractor. We processed independently each 30-second segment to generate intermediate representations, which were then concatenated per subject. These subject-level feature sequences were passed to a secondary transformer encoder, specifically designed to model long-range temporal dependencies and capture contextual relationships across segments. This architecture aimed to capture the sequential structure of a participant’s full recording, replacing the prior ensemble-per-patient approach. Finally, to better understand the internal representations learned by Whisper’s encoder and identify the most important audio features contributing to its predictions, we conducted an analysis using acoustic probing feature analysis across layers to determine the importance of some acoustic attributes learned in the models’ layers. These analyses provide insight into the hierarchical feature representations developed by the encoder and offer interpretability into how the model distinguishes between cognitive states.

Data

To expand the training corpus, we accessed and downloaded the following studies from the DementiaBank repository: Baycrest, Delaware, Holland, Kempler, Lanzi, Lu, Pitt, VAS, WLS, DePaul, Hopkins, Jalvingh, Dem@Care, Chou, Ye, and Ivanova. For recordings involving multiple speakers, we leveraged the available transcripts to identify speaker-specific time intervals. This identification enabled us to isolate and retain only the segments corresponding to the participant, excluding those of the interviewer or other voices. Then, the audio files were trimmed accordingly. Additionally, we excluded some studies following a quality or content-based review. Dem@Care was removed because the transcripts were generated using automatic speech recognition (ASR), making it unclear who was speaking and whether an interviewer was present. Lu (Taiwanese) was excluded due to unlinked transcripts, which prevented reliable alignment between audio and text. All retained audios were processed using the Phase 2 audio preprocessing pipeline with minor changes and segmented into 30-second chunks, in alignment with Whisper’s encoder. Additionally, from each recording, we extracted 90 linguistic features every 0.2 seconds using a sliding window with a 0.1-second overlap. To reduce input dimensionality and improve computational efficiency, these features were averaged over 1-second intervals. Where available, we extracted demographic information, including gender, age, and level of education from transcripts or metadata files, for potential use in downstream modeling or stratification analyses. The diagnoses were organized into four classes: (i) healthy controls (HC; 2044 samples), (ii) mild cognitive impairment (MCI; 559 samples), (iii) Alzheimer’s disease (AD; 946 samples), and (iv) other dementias (OD; 171 samples). The last class comprises participants with the following diagnoses across all the studies: dementia, vascular, PPA-nos, LVPPA, SVPPA, NFAPPA, FTD, semantic PPA, Pick’s, aphasia, SD, LPA, PNFA, and other. The final curated dataset comprises recordings in four languages, English, Mandarin, Spanish, and German, spanning 9 distinct speech tasks, 3,720 participants, and 16 individual studies. For architecture optimization, the dataset was partitioned into 2,817 training samples and 903 test samples. The test set was deliberately constructed to evaluate both in-distribution and out-of-distribution generalization. It included all test data from Phase 2 of the Eureka challenge, allowing for a direct comparison of performance

improvements. Additionally, it incorporated all available recordings from the Alexa task, which was not used for training, to assess out-of-distribution generalization across tasks. And, to evaluate language-based generalization, we also included all Mandarin-based studies. The remainder of the test set was selected to maintain an approximate 80/20 train-test split. A more comprehensive and rigorous generalization framework, designed to assess cross-task and cross-language performance, was implemented after the architecture was finalized (see Generalizability).

Our best-performing model, selected for its balance between classification accuracy and generalizability, was a fine-tuned version of the Whisper Large encoder, followed by an MLP that served as the classification head. The final prediction for each participant was computed using a segment-level ensemble approach. Specifically, we averaged the class probabilities generated from each 30-second audio segment to obtain the participant-level prediction. Compared to our Phase 2 submission, which achieved a log-loss of 0.6976, the final model, benefiting from expanded training data, selective layer freezing, and participant-level ensemble, achieved a substantially improved log-loss of 0.4680. On the full test set, we evaluated model performance by class using both accuracy and area under the precision-recall curve (AUPRC). Accuracy scores were 0.7574 for HC, 0.6777 for MCI, 0.7286 for AD, and 0.8992 for OD. Corresponding AUPRC values were 0.799 for HC, 0.440 for MCI, 0.798 for AD, and 0.399 for OD.

These results indicate high performance in distinguishing AD and HC, with more modest performance in identifying MCI and OD. The relatively lower performance in these latter groups can be partially attributed to class imbalance and linguistic variation. The dataset is predominantly composed of English recordings, with very limited samples in German and Mandarin, and a significant skew toward HC and AD cases. Furthermore, the OD category is a heterogeneous set of clinical conditions that may present overlapping speech characteristics with other groups, making accurate classification even more challenging. The model's comparatively stronger performance on AD and HC highlights the robustness of the approach in detecting more distinct cognitive profiles, while also underscoring the need for more balanced and granular data, particularly for intermediate and atypical dementia presentations. A more comprehensive analysis of each subset and the model's generalizability can be found in the key insights document.

Generalizability

To evaluate the generalizability of our model, we pursued two complementary strategies: (1) the proposal of a new generalization benchmark, and (2) the use of speech-to-speech translation. The latter approach addresses the linguistic imbalance in available datasets, which are predominantly in English. To mitigate this bias while leveraging the strong performance of English-language models, we designed a speech translation pipeline that aims to convert non-English recordings into English, preserving the key acoustic characteristics of the original speakers. The pipeline is built entirely on open-source models. Firstly, non-English recordings were transcribed using CrisperWhisper, a tool capable of detecting linguistic features relevant to cognitive decline, such as word repetitions, partial words, and filler expressions. Then, we evaluated four machine translation approaches: Whisper¹, SeamlessM4T², Llama-3.2³, and MarianMT⁴. Translated transcripts were used to synthesize English audios using OpenVoicev2, which enables the preservation of speaker's vocal traits, thereby maintaining acoustic fidelity.

While speech-to-speech translation is considered a largely solved problem in the AI community, primarily due to the use of closed-source systems, our evaluation revealed that currently available open-source models are not yet reliable enough to establish a fully automated pipeline without human oversight. Key limitations included truncated outputs for longer recordings, the insertion of filler words not present in the original audio, and, in some cases, complete translation outputs dominated by filler content or empty. Across all evaluated systems, transcriptions tended to be more accurate than the resulting translations. Despite these limitations, we believe this direction holds significant promise. Therefore, we identified a subset of audios that were accurately translated and applied post-processing steps to retain only the translated content, removing any hallucinated or extraneous output unrelated to the original speech. Then, we conducted a pilot evaluation to assess their utility in downstream prediction. As demonstrated in the Insights document, the results were highly promising. Regarding the first strategy (development of a new generalization benchmark), we propose a data partitioning scheme, training methodology, and set of

evaluation metrics designed to enable a more rigorous and comprehensive assessment of model robustness. To move beyond the limitations of log loss alone, we propose incorporating both accuracy and precision-recall (PR) curves in the evaluation protocol. While log loss provides insight into model calibration, PR curves offer a more informative view in class-imbalanced scenarios, allowing us to evaluate the trade-off between sensitivity (recall) and the ability to avoid false positives (precision). This information is critical in ADRD screening, where the cost of misclassification varies by clinical context and diagnosis stage.

To assess the model's sensitivity to language and task variability, we designed two leave-one-out generalization protocols. The first one focuses on linguistic robustness. We trained three separate models, each excluding all samples from one of the target languages (English, Spanish, and Mandarin) and then evaluated their performance on the held-out language. This setting allows us to measure how well the model generalizes to entirely unseen linguistic distributions. The second protocol targets task-level generalization. Given that some audio recordings may include overlapping elements from multiple cognitive tasks, and that task-level segmentation is not automated, we selected three representative tasks (Image Description, Word Fluency, and Discourse tasks) and manually curated a subset of recordings in which each task was predominantly featured. While some recordings may still contain incidental elements of the excluded task (e.g., a participant in a Discourse task may describe an image), the majority of content related to the held-out task was removed. Then, we trained one model per task using a leave-one-task-out scheme, excluding all recordings where the selected task was a primary component. In total, three models were trained and evaluated in this setting. This design enables us to quantify the contribution of each task to model performance and identify which tasks are most sensitive to omission during training. Together, these generalization experiments provide a more comprehensive understanding of model behavior under realistic deployment conditions and offer insights into failure modes not observable under standard cross-validation.

Methodology

Throughout Phase 3, we conducted a comprehensive series of experiments to improve model performance through architectural refinements, and metadata integration. We began by establishing a Phase 2 performance baseline using the distilled Whisper model on the first 30 seconds of each audio sample. Then, we extended this approach to full-audio training by dividing each recording into 30-second segments and aggregating predictions across segments to generate participant-level classifications via an ensemble. Building on these initial experiments, we transitioned to a more complex model, Whisper Large, to evaluate whether increased model capacity would yield performance gains. Given the model's size and the limited quantity of training data, we applied selective layer freezing, retaining only the first two and final encoder layers as trainable. This technique was tested across multiple variants, including Whisper Large, distilled Whisper, and Crisper Whisper, aiming to preserve the benefits of pretraining while allowing for targeted fine-tuning. We also explored the integration of demographic metadata, specifically age, gender, and education using three strategies: (i) a late-fusion approach with a multilayer perceptron, (ii) adding a learnable embedding directly into the encoder (analogous to positional encoding), and (iii) incorporating a learnable scaling factor to modulate the embedding's influence during training.

Additionally, due to the diverse length distribution of the audio recordings, we developed a two-stage architecture. In the first stage, we extracted segment-level features using the best-performing Whisper model. These features were then fed into a secondary transformer designed to model long-range sequential dependencies across segments. To support this, we implemented a modified positional embedding scheme specifically designed for speech sequences. We iteratively refined this architecture by integrating additional metadata (e.g., age and gender), including acoustic features from Phase 2, tuning hyperparameters, and modifying the loss function to better account for severe class imbalance. Additionally, we experimented with reducing the segment duration from 30 seconds to 10 seconds to obtain more granularity and context within the sequence model. Following all experimentation, we observed that the best-performing model varied by diagnostic category. For example, the highest performance for healthy controls was achieved with the two-stage transformer on 30-second segments. For MCI and AD cases, Distilled Whisper outperformed other models. For other types of dementia, the two-stage transformer on 10-second segments was most effective. Nevertheless, the model with the strongest overall performance across all classes was Whisper Large.

References

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